PRE-ALGEBRA (7TH) · UNIT 9: SPATIAL THINKING

Skate Park Designer

Use angles, polygons, circles, and transformations to design a town skate park.

Honors

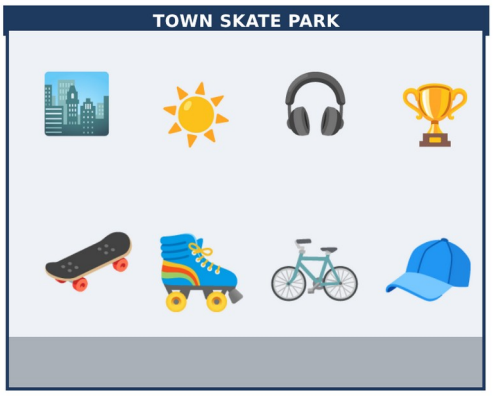
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The town council approved a new skate park and wants students to design it. The ramps need exact angles, the rails run parallel, the deck is a polygon, and the bowl is a circle.

You will use angle relationships, polygons, circles, and transformations to create a blueprint and a logo for the park.



Ramp Rule

Angle of ramp

15° to 35°

Bowl

Circle, diameter

24 ft

Learning Goals

- I can justify angle relationships with algebra and reasoning.
- I can use polygon properties to design tiling and decks.
- I can apply circles and compositions of transformations.

Project Timeline

When	What you will do
Day 1	Angle Proofs
Day 2	Deck Design
Day 3	Bowl & Logo
Final	Skate Park Blueprint + reflection

Part 1: Angle Proofs

Day 1 · Rails are parallel; ramps meet the 15°–35° rule.

Feature Cards

Quarter Pipe	Fun Box	Rail Pair
Ramp angle 30°	Regular hexagon top	Two parallel rails cut by a path

Bowl	Pyramid	Logo
Circle, d = 24 ft	Square base, 4 triangle sides	Must have rotational symmetry

1. A path crosses two parallel rails. One angle is $(x^2 - 10)°$ and its corresponding angle is $(3x + 30)°$ with $x > 0$. Find x and the angle.

2. Prove that same-side interior angles are supplementary using a corresponding angle and a linear pair.

3. A ramp rises 4 ft for each 10 ft forward. Use a protractor drawing to estimate the angle. Does it meet the rule?

4. The angles of a triangular kicker are in the ratio 1 : 2 : 6. Find them. Does the smallest meet the ramp rule?

Part 2: Deck Design

Day 2 · The deck is tiled with polygons.

5. Which regular polygons tile a flat deck by themselves? Use interior angles to explain.

6. Can regular octagons and squares tile together? Show the angle sum at a vertex.

7. A regular polygon deck has interior angles of 150°. How many sides? What is its angle sum?

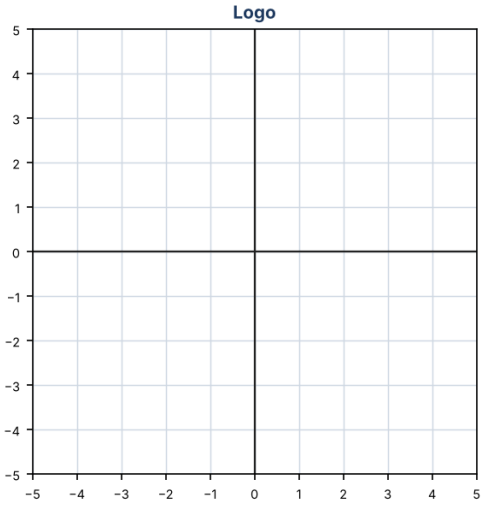
8. Explain why every regular polygon has exterior angles that add to 360°.

Part 3: Bowl & Logo

Day 3 · Use π.

9. The bowl is a circle with diameter 24 ft. A safety zone 4 ft wide surrounds it. Find the area of the safety zone in terms of π and to the nearest square foot.

10. Reflect the logo triangle (1, 1), (4, 1), (1, 3) across the y-axis, then rotate the image 180° about the origin. What single transformation does the same thing?



11. Design a logo with exactly 4 lines of symmetry and rotational symmetry of order 4. Explain how you know.

Final Product: Skate Park Blueprint

Your final product must include:

- Blueprint with angle proofs
- Deck tiling plan with angle sums
- Bowl and safety zone areas
- Composite transformation of the logo
- Presentation to the town council

Reflection

R1. Why do only some regular polygons tile a plane?

R2. How can two transformations combine to equal a single transformation?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Angles, polygons, circle measures, and transformations are correct and produce a buildable design. Goes beyond the minimum with a creative, realistic choice.	Angles, polygons, circle measures, and transformations are correct and produce a buildable design.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Blueprint is labeled with angles and measurements and includes the logo and calculations. Polished, organized, and visually engaging.	Blueprint is labeled with angles and measurements and includes the logo and calculations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.